AP PRECALCULUS

Unit 1 • Lesson 1.7 notes

Lesson 1.7 • Rational Functions and End Behavior

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Determine the end behavior of a rational function by comparing degrees.
- Find horizontal and slant asymptotes.

Vocabulary. rational function, horizontal asymptote, slant (oblique) asymptote, polynomial division.

1. Quotient of leading terms

For $r(x) = \dfrac{p(x)}{q(x)}$, as $|x| \to \infty$ the function behaves like $\dfrac{\text{leading term of } p}{\text{leading term of } q}$.

Key idea

$\deg p < \deg q$	horizontal asymptote $y = 0$
$\deg p = \deg q$	horizontal asymptote $y = \frac{\text{leading coefficient of } p}{\text{leading coefficient of } q}$
$\deg p = \deg q + 1$	slant asymptote: the quotient of $p \div q$
$\deg p > \deg q + 1$	behaves like a polynomial; no linear asymptote

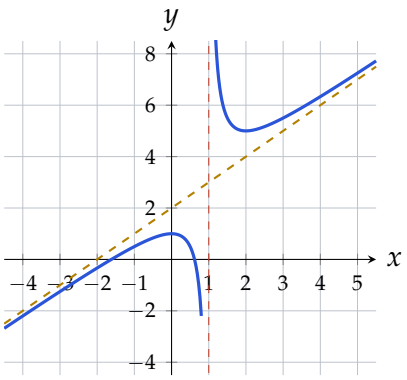
Example 1 • Horizontal asymptotes

Solution. $\dfrac{3x+1}{x^2-4}$: $y = 0$. $\dfrac{8x^2-x}{2x^2+5}$: $y = 4$. $\dfrac{x^3}{x^2+1}$: behaves like x, so it goes to $\pm\infty$ (and in fact has the slant asymptote $y = x$).

Example 2 • Slant asymptote

Find the slant asymptote of $s(x) = \dfrac{x^2+x-1}{x-1}$.

Solution. Divide: $x^2+x-1 = (x-1)(x+2)+1$, so $s(x) = x+2+\dfrac{1}{x-1}$. As $x \to \pm\infty$, $\frac{1}{x-1} \to 0$: the slant asymptote is $y = x+2$.



Common mistakes

- Reading the asymptote from the constant terms instead of the leading coefficients.
- Forgetting that a graph *can* cross a horizontal asymptote for moderate values of x; the asymp-

tote describes the ends only.

Quick check

1. Horizontal asymptote of $\dfrac{5-2x^3}{x^3+x}$? 2. Slant asymptote of $\dfrac{2x^2+3x}{x+1}$?

Answers. 1. $y=-2$. 2. $2x^2+3x=(x+1)(2x+1)-1$, so $y=2x+1$.